Information

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Team F:
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ILR08

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Individual Progress

Since the previous lab, I have been working with Dillon regarding the connector piece. We would 3d print our first attempt to allow the legs to disconnect with the main central base of the hexapod, allowing us to fit within the 3 foot by 3 foot constraints when packaged up.

With our team still not getting our servo heads in time for this check off, I would work with Dillon on pivoting to creating a moving mock up system, mainly to allow our programmers to have a system to push IK, whether from controller or later Rasberry Pi + ESP32, and test coding as we wait for the servo heads to create the 1:1 leg. I would 3d print and assemble the arm and later redo the power systems to not burn out the smaller servos and still run safely with the same NIMH batteries.

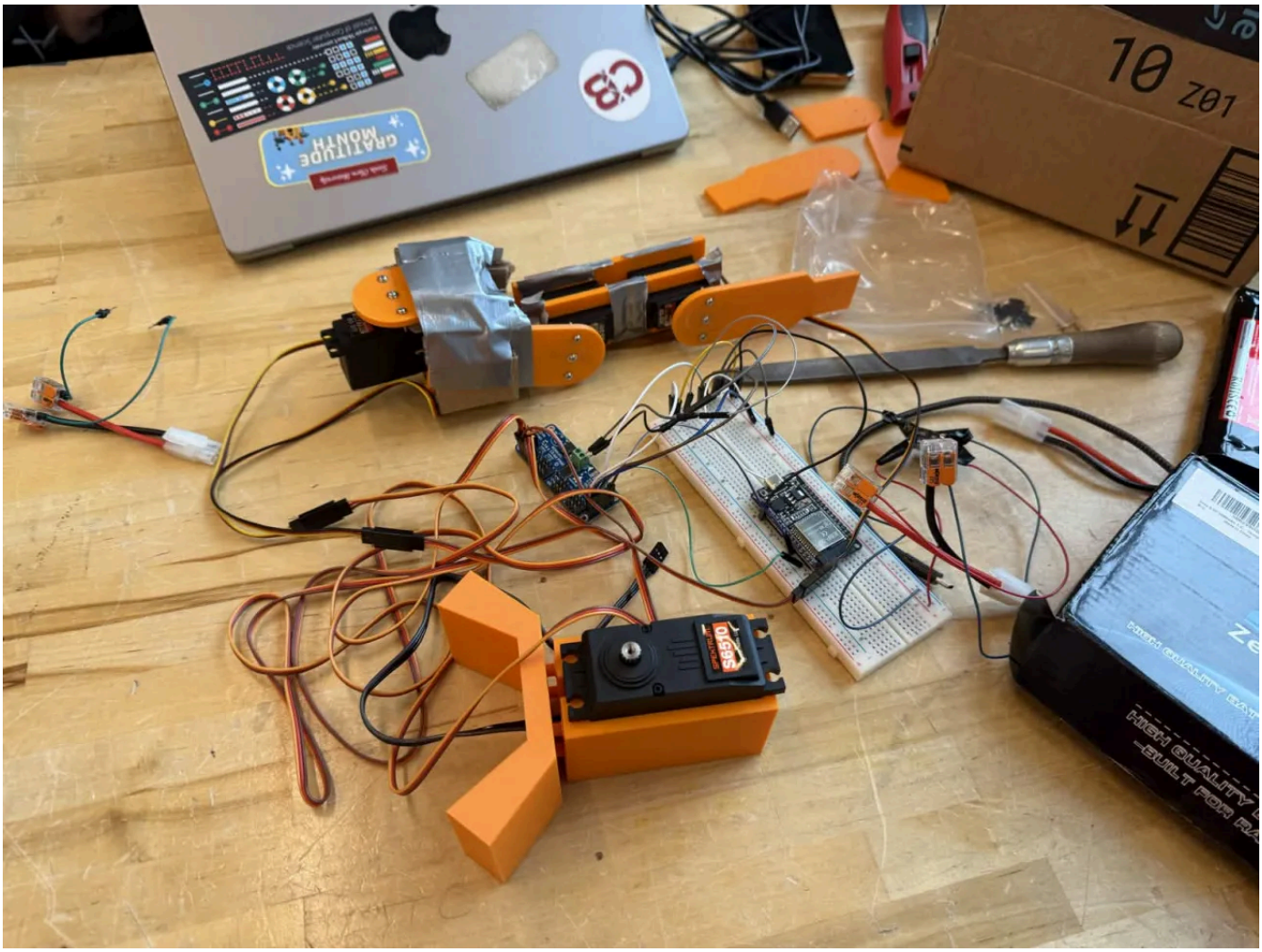

Figure 1: Mock Up Leg Created

Challenges

There were multiple challenges that arose. The first issue was regarding shipping. Due to the metal servo horns/heads still not arriving, we had to pivot due a fall back to create some system to allow the software team to test IK. However, since the servo heads just arrived after class, we should soon be able to create our actual leg for testing

Another issue that would arise would be with our leg connector piece. We would try to connect the two 3d printed pieces, and due to both tolerances and stress, we would go from having issues with pieces sliding into each other to pieces snapping under stress.

One more issue that would arise was with wiring. While in storage, some wires would get jumbled, and while it was an issue easily fixed, it did re-raise the issue of wiring and having to both keep wire stable alongside labeled.

Teamwork

While I was working:

Dillon was working on the CAD. We would work to CAD then make the servo connector piece alongside the scaled down leg to test IK

Leena would create the code to test controlling each of the three servos in our scaled down leg.

Thomas would work with the Raspberry Pi, allowing for OpenCV control with detecting and reading april tags. Thomas would also work on allowing anyone on CMU Wifi to SSH into the Rasberry Pi, so all of us can check from our laptop on what the camera sees.

Plans

With us finally getting some of the metal servo heads/horns, me and Dillon will be working on finally creating V2 of the leg with the right servo head. We will then quickly finalize the design and do changes before mass producing each leg. We will also be working on the chassis in parallel, working on the chassis+chair and reducing the friction of the chair with teflon tape.